pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

Retrieval Evaluation: Recall and MRR

WEEK 3 · DAY 1

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Today: scoring retrieval on its own

A labelled set, no model for answers

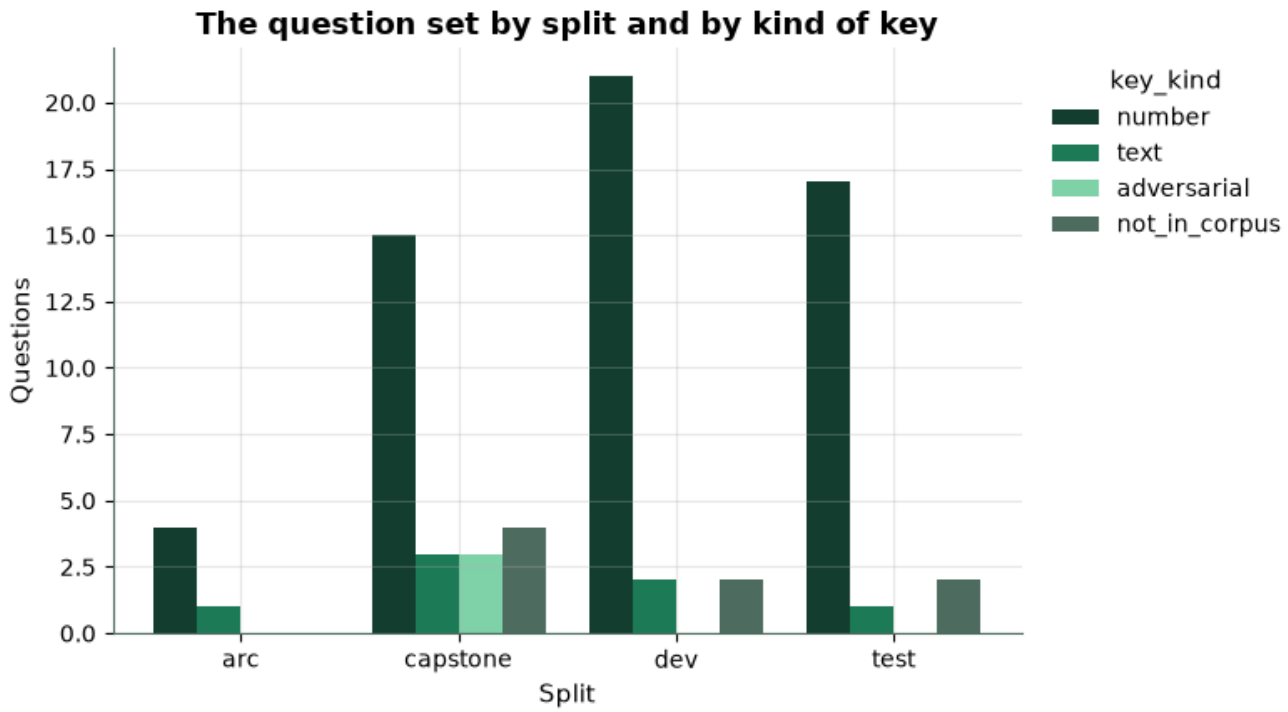
Chunks	1,158 at 512/64
Questions	25 development, 20 test
Scored	23 and 18 (a page in the key)
Pipelines	MiniLM, BM25, hybrid, + reranker
Requests	none; no key

Today

1. Score retrieval before answers
2. The labelled set
3. evaluate, recall at k, MRR
4. By item: where retrieval misses
5. A miss counted as zero
6. The test questions, once, with an interval
7. What recall does not say

Name the rule of every count; choose nothing today; open the test rows once.

Written from pages, never from chunks



One row holds

id, split, question, answer_key

key_kind: number, text, not_in_corpus, adversarial

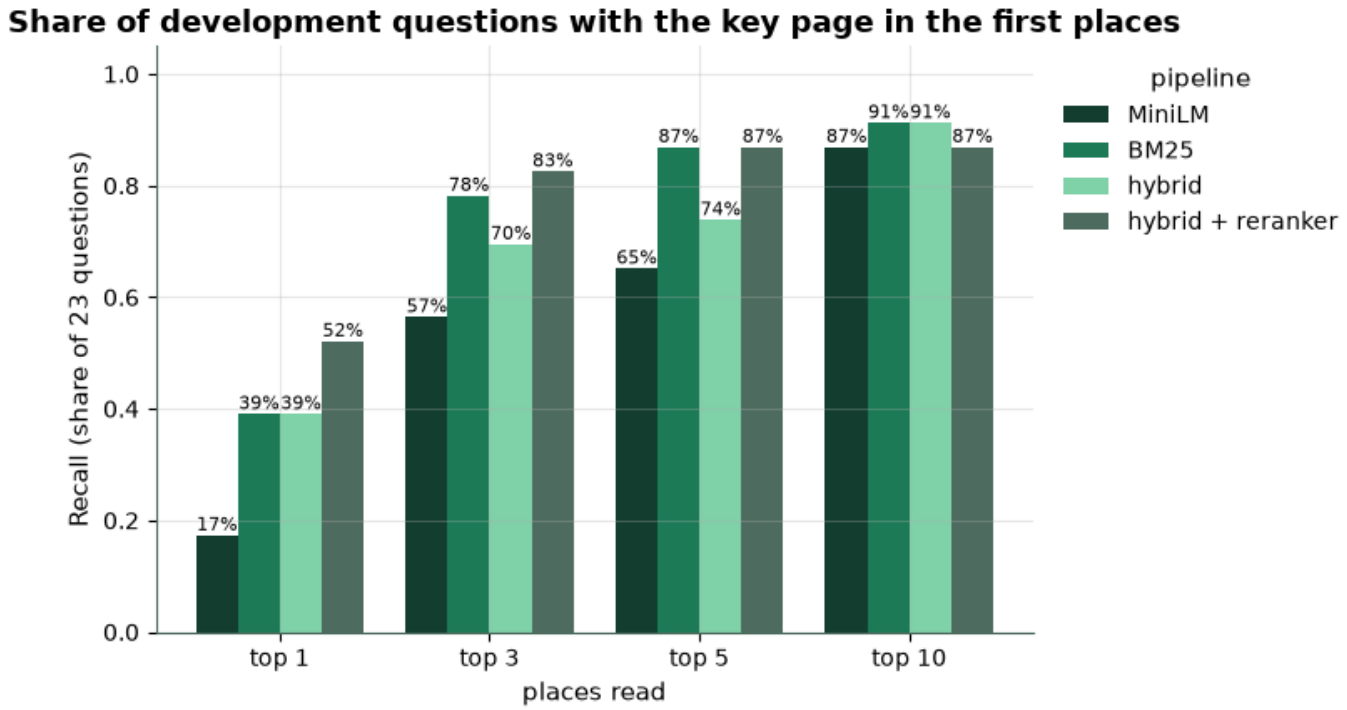
expected_item, expected_page

key_quote and written_from (a page)

notes

arc 5, capstone 25, dev 25, test 20. 8 of 25 development rows name another place for the figure in their notes (dev01, dev02, dev04, dev09, dev15, dev19, dev20, dev22): the key has one page.

Four pipelines on the development questions



Recall at 1, 3, 5, 10 and MRR

MiniLM

`0.17  0.57  0.65  0.87  MRR 0.40`

BM25

`0.39  0.78  0.87  0.91  MRR 0.61`

hybrid

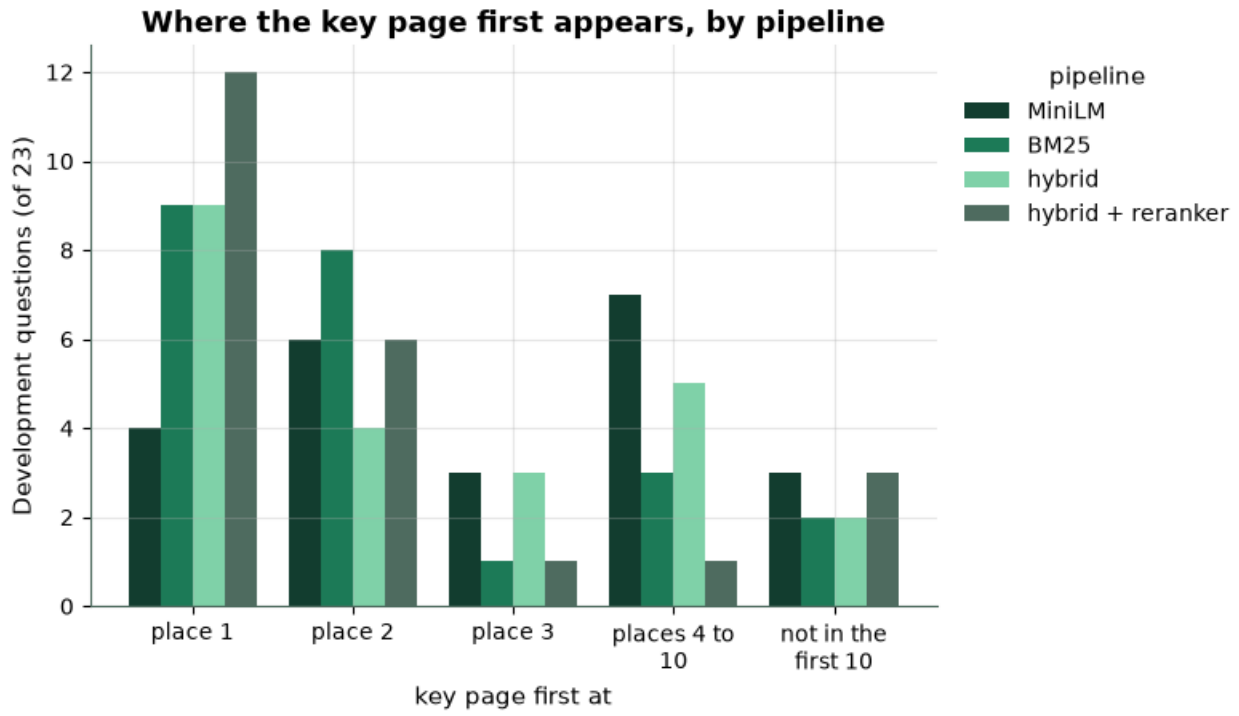
`0.39  0.70  0.74  0.91  MRR 0.56`

hybrid + reranker

`0.52  0.83  0.87  0.87  MRR 0.68`

By the page rule. A miss counts 0 and stays in the mean. 23 questions: one question is 0.04.

Where the key page first appears

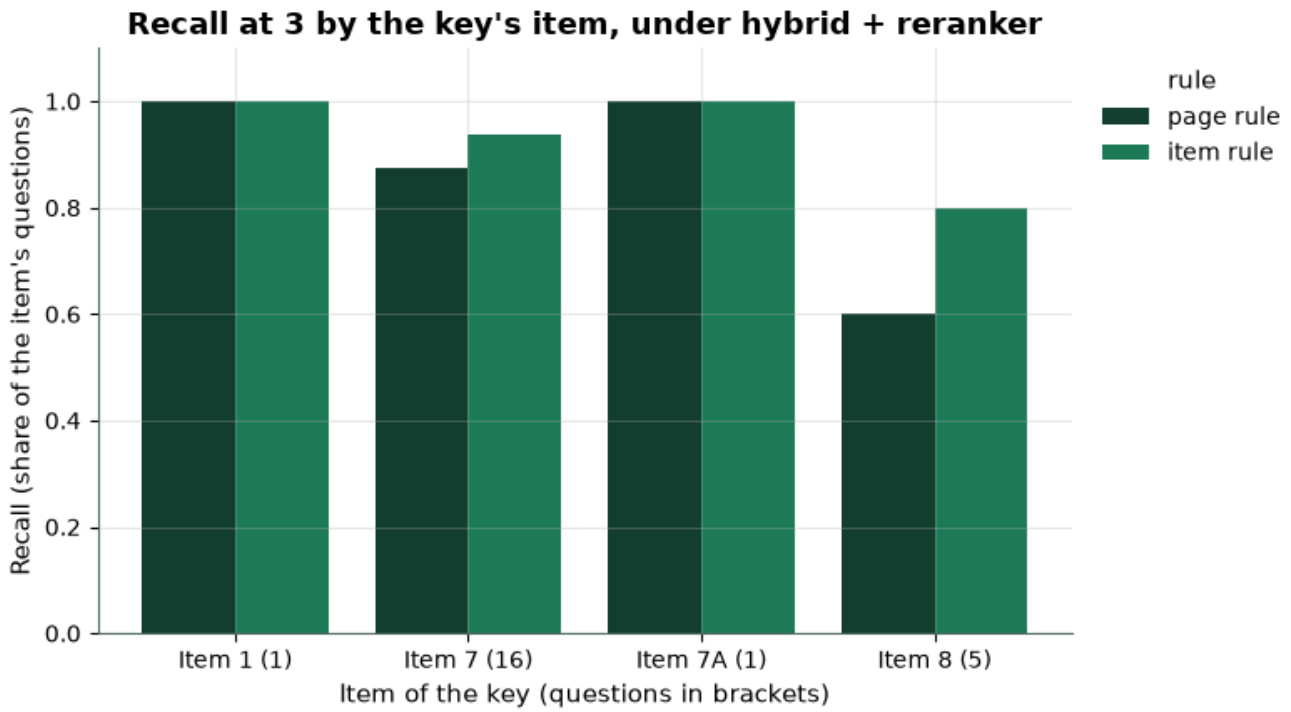


Questions, by first place

	1	2	3	4-10	none
MiniLM	4	6	3	7	3
BM25	9	8	1	3	2
hybrid	9	4	3	5	2
hybrid + reranker	12	6	1	1	3

Recall at 10 says the page is somewhere in ten places; MRR says how early.

The page rule and the item rule



At 3: page rule 19 of 23, item rule 21

item	n	page	item
1	1	1	1
7	16	14	15
7A	1	1	1
8	5	3	4

The item rule counts any chunk from the key's item. Groups this small describe; they do not estimate.

The four misses at 3

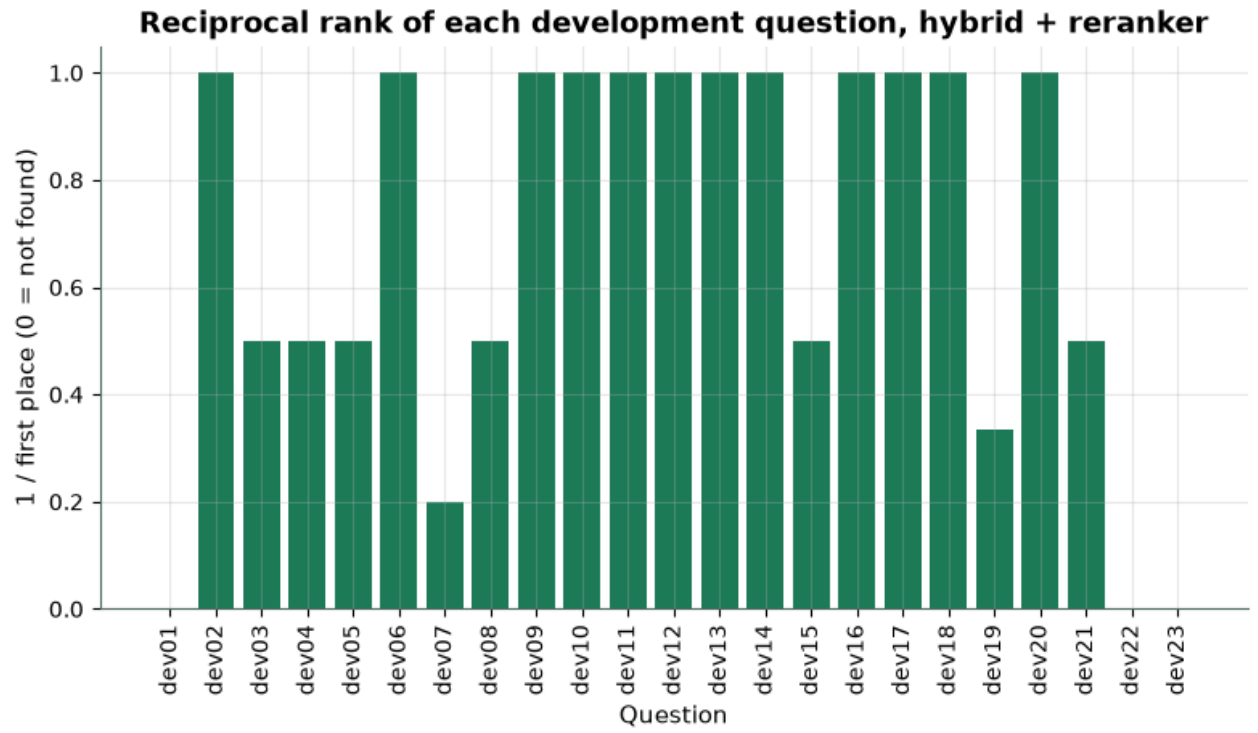
Place of the first chunk on the key page (- = not in the first 10)

id	item	key page	by page	by item	pages of the first 3
dev01	7	37	-	1	40 78 36
dev07	7	38	5	5	76 101 76
dev22	8	52	-	2	43 61 94
dev23	8	98	-	4	34 34 34

dev22: total assets, key on page 52 (Item 8)

The first place is an Item 7 paragraph that uses the question's words and holds no balance sheet figure; the chunk with the key is the balance sheet. Share of lines that look like table rows: first place 0.25, chunk with the key 0.70. The notebook does not test why the cross-encoder scored the paragraph above the table.

Reciprocal rank, with the misses at 0



Two means of the same run

0.6754

MRR over all 23 questions, misses counting 0

0.7767

mean over only the 20 questions that were found

3

misses at 0: dev01, dev22, dev23

A mean over the questions that were found describes the retriever where it is already good.

Opened once, for the setting chosen before

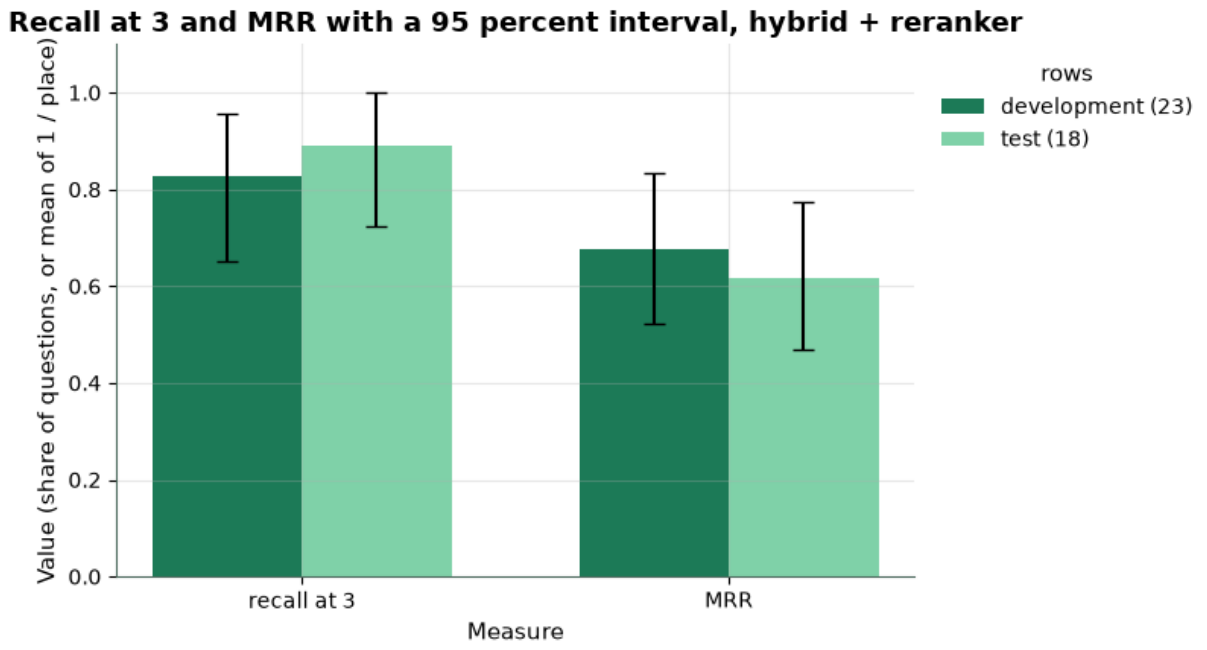
hybrid + reranker, by the page rule

rows	recall 1	recall 3	recall 5	recall 10	MRR
development (23)	0.52	0.83	0.87	0.87	0.68
test (18)	0.39	0.89	0.94	0.94	0.62

The one cell that reads the test rows

recall at 1 7 of 18 (0.3889), recall at 3 16 of 18 (0.8889), recall at 5 17 of 18 (0.9444), recall at 10 17 of 18 (0.9444); MRR 0.6157. Not in the first 3: test01, test08. If the number disappoints, the setting stays and the number is reported.

An interval, and which function made it



ensemblekit and plain resampling

development, recall at 3 0.83

```
ens 0.65-0.96   plain 0.65-0.96
```

development, MRR 0.68

```
ens 0.52-0.83   plain 0.52-0.83
```

test, recall at 3 0.89

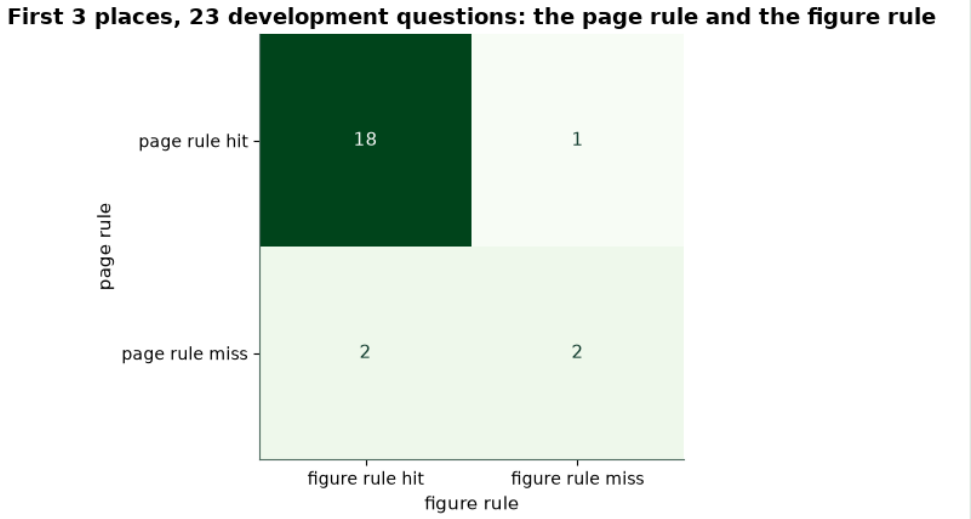
```
ens 0.72-0.94   plain 0.72-1.00
```

test, MRR 0.62

```
ens 0.46-0.76   plain 0.47-0.77
```

ensemblekit redraws a resample with one class only; a resample of 18 with no miss at 3 has chance 0.120. Say which function made the interval.

The page rule and the figure rule



First 3 places, 23 development questions

page rule and figure rule both hit 18; page rule only 1 (dev19); figure rule only 2 (dev01, dev07); neither 2 (dev22, dev23)

dev19: the page is returned, the figure sits in the neighbour chunk

dev01 and dev07: the figure is in the chunks, from another page

A page rule undercounts where a figure has two homes

Recall stops at the model's door

Neither rule is wrong. Each measures something else, and a count must say which.

Excel side by side, and Git

Excel side by side (tested in Excel)

`=IFERROR(MATCH($B2,$C2:$L2,0),0)`	first place of the key page; equal to evaluate's first_rank for 23 rows
`=ISNUMBER(MATCH($B2,$C2:$E2,0))`	hit at 3, equal to Python for 23 rows
`=COUNTIF(O2:O24,TRUE)/ROWS(O2:O24)`	recall at 3: 0.8261, equal to Python
`=SUMPRODUCT((N2:N24>0)*(N2:N24<=5))/ROWS(N2:N24)`	recall at 5: 0.8696, equal to Python
`=IFERROR(1/MATCH($B2,$C2:$L2,0),0)`	reciprocal rank; the miss becomes 0, equal to Python for 23 rows
`=AVERAGE(P2:P24)`	MRR: 0.6754, equal to Python
`=AVERAGEIF(P2:P24,">0")`	leaves the misses out: 0.7767, the buggy function's number
`=COUNTIFS($M$2:$M$24,"8",$O$2:$O$24,TRUE)/COUNTIF($M$2:$M$24,"8")`	Item 8 recall at 3: 0.6000, equal to Python
`=PERCENTILE.INC(A1:A1000,0.025)`	2.5th and 97.5th percentiles of 1,000 means: 0.7222 and 1.0000
`=ISNUMBER(MATCH(T1,$C3:$E3,0))`	FALSE when the key page is stored as text, TRUE as a number

Git: retrieval_eval.csv, 23 development rows, then 18 test rows; 53 tests.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the filter first_rank > 0 drops every question that was not found, so the mean ignores the misses.

Code that runs is not the same as code that is right.

What you can do now

- Score retrieval before answers: a key that names a place
- evaluate, recall at k, MRR; a miss counts 0
- Read by item; name the rule of every count
- Open the test rows once; say which function made the interval
- A page-rule hit is not a figure in the chunks, and not a right answer

Week 3, Day 2 LLM judges and their limits